

Sieve Estimator for Function Autoregressive HMM - A First Attempt

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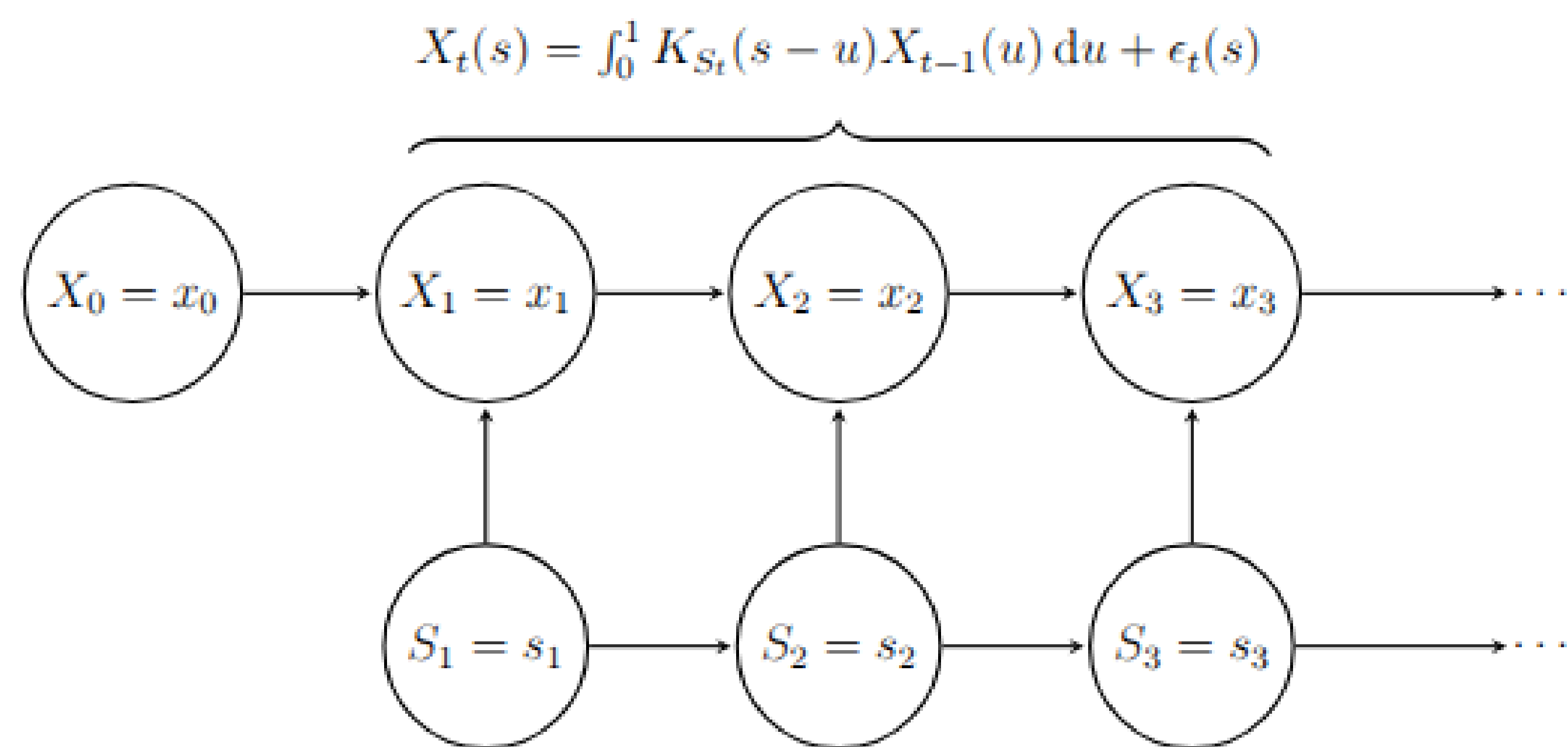
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Introduction

Many real-world functional time series switch between latent regimes that share similar average behavior but differ sharply in how strongly they depend on their own past. Standard hidden Markov models struggle to detect this kind of regime when the means look alike, because they only see marginal shape, not dynamics. We introduce the Sieve-Estimated Functional Autoregressive Hidden Markov Model (SI-FAR-HMM), which embeds a sieve-estimated FAR(1) operator inside each hidden state, allowing the model to distinguish regimes by their autoregressive persistence rather than their level alone.

Model Diagram



The model is similar to a HMM; only now with dependencies between observations.

What It Can and Cannot Do

Where SI-FAR-HMM thrives:

- Detecting regime changes driven by dynamics not level.
- Closed form tractable solution and estimation of FAR process.

Where SI-FAR-HMM has trouble:

- Sieve order is capped by data.
- Model initialization; not unique to SI-FAR-HMM but always worth highlighting.

Where Do Functional Observations Live?

Functional observations live in a Hilbert space, specifically $L^2([0, 1])$. This space has an inner product $\langle x, y \rangle = \int_0^1 x(t)y(t)dt$.

A random curve $X = \{X(t), t \in [0, 1]\}$ is an element of L^2 equipped with the Borel σ -algebra. We can define the mean as:

$$X \text{ integrable if } E[\|X\|] = E\left[\int_0^1 X^2(t)dt\right]^{1/2} < \infty$$

If X integrable then $\exists!$ function μ s.t. $E[\langle y, X \rangle] = \langle y, \mu \rangle \forall y \in L^2([0, 1])$.

Typically we assume $EX = 0$ and square integrable. If this is the case then we can define the covariance operator of X as:

$$C(y) = E[\langle X, y \rangle X], \quad y \in L^2([0, 1]).$$

From here we can create functional analogues of nearly all of the usual statistics objects and theorems, like finding $\hat{\mu}$ and a functional C.L.T. But be wary, these objects have infinite dimension and quickly become complicated to work with^{a!}

^aQ: Is there a way to invert a matrix with infinite dimensions?

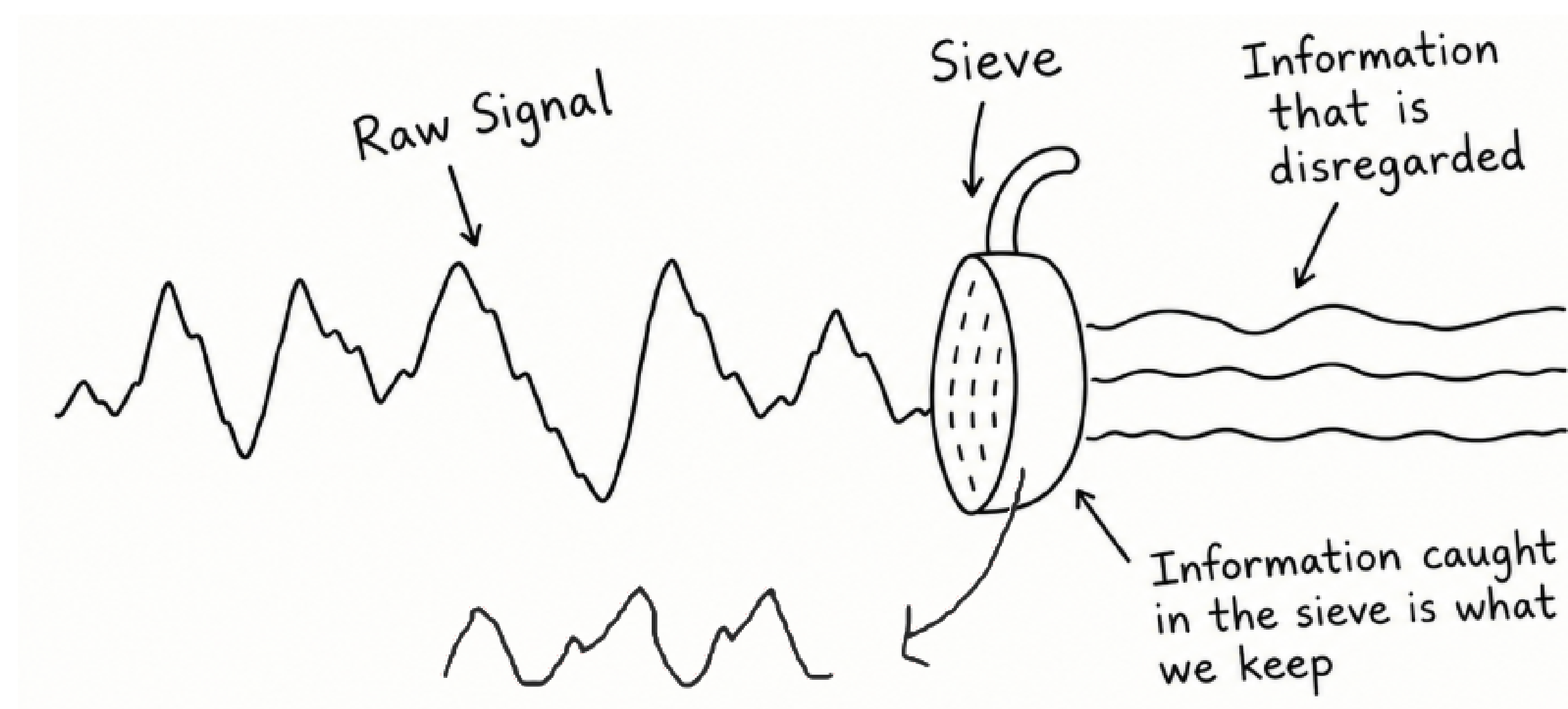
A Sieve

Developed by [2], a sieve is a powerful technique where the optimization over an infinite dimensional parameter space Θ takes place over a finite dimensional subset $\Theta_m \subset \Theta$.

In the context of this work we define the sieves Θ_m as the finite Fourier transformation of elements in Θ_m :

$$\phi_0 = \mathbb{1}_{[0,1]}, \quad \phi_{2k-1}(t) = \sqrt{2} \cos(2\pi kt), \quad \phi_{2k}(t) = \sqrt{2} \sin(2\pi kt), \quad 1 \leq k \leq m. \quad (1)$$

You can think of a sieve as a pasta strainer. Information is passed through a sieve and we keep only what is caught. How much information we keep is determined with how fine we make the pasta strainer.



As the density of the observations over a fixed grid increases we can increase the fineness of the sieve for a more precise estimation.

Functional Autoregression and Sieves (SI-FAR)

The work of [1], develops a procedure to estimate a FAR process $X_t = \rho(X_{t-1}) + \epsilon_t$ using a sieve. Note that:

1. $\{X_t\}$, $t = 1, \dots, T$ is strictly stationary.
2. $\rho : \mathcal{H} \rightarrow \mathcal{H}$ is a bounded linear operator.
3. (ϵ_t) is strong $\mathcal{H}$ -white noise.

The operator ρ is assumed to be a convolution operator with 1-periodic kernel $K \in L^2([0, 1])$ satisfying $\|K\|_{L^2} < 1$, so that

$$\rho(f)(t) = \int_0^1 K(t-s)f(s)ds.$$

Define the sieve Θ_m by

$$\Theta_m = \left\{ K \in L^2([0, 1]) \left| K(t) = c_0 \mathbb{1}_{[0,1]}(t) + \sum_{k=1}^m (c_k \cos(2\pi kt) + d_k \sin(2\pi kt)) ; \sum_{k=1}^m k^2 (c_k^2 + d_k^2) \leq m \right. \right\}, \quad (2)$$

and in [1], closed form solutions for the sieve parameters are derived.

SI-FAR-HMM In a Nutshell

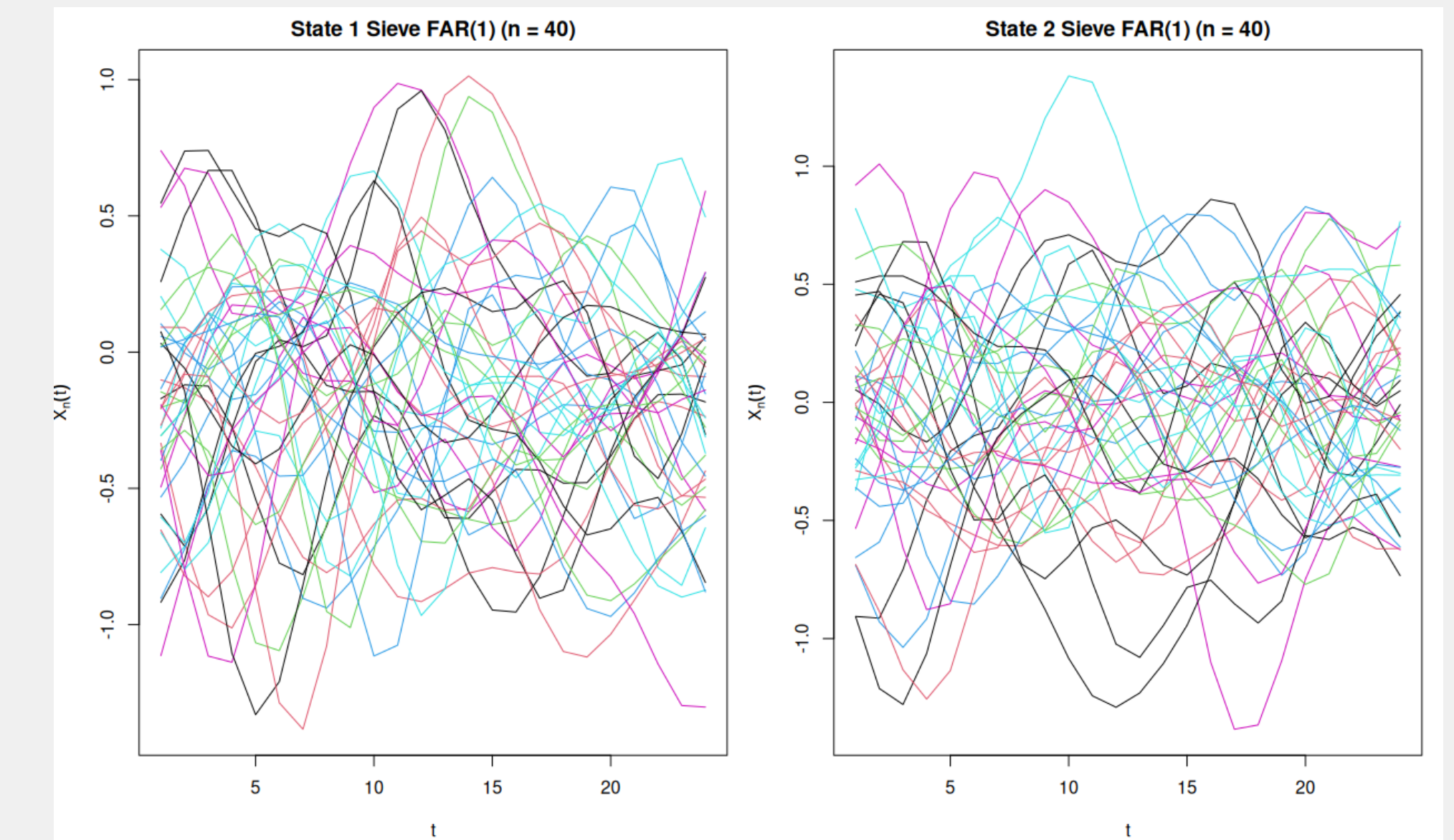
We can now assemble the pieces of the puzzle. Let the SI-FAR-HMM have the following parameters:

- Number of hidden states J . Dimension of sieves m for all states $j \in 1, \dots, J$.
- Transition probability matrix Γ with stationary probability π .
- A FAR process for each state with kernel K_j estimated by the sieve $\hat{\Theta}_{m,j}$.

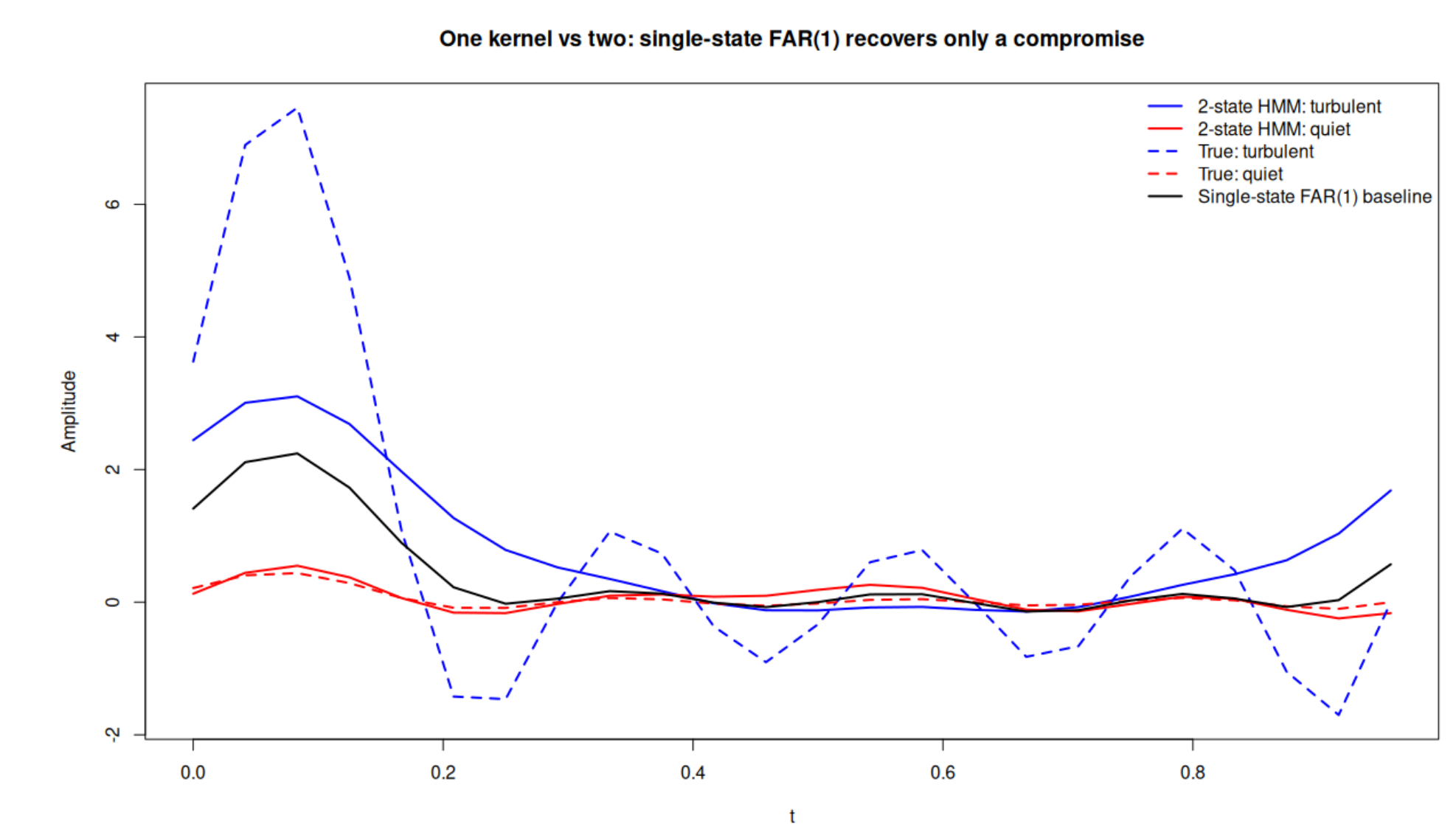
We can then maximize the SI-FAR-HMM using the conventional Expectation Maximization Algorithm.

A Data Simulation

Both states share the same stationary mean (≈ 0) and variance, but differ sharply in temporal persistence: State 1 a turbulent state where each curve strongly echoes the last, and State 2 a quiet state where curves are nearly independent draws. Below is 40 consecutive observations from each state.



Versus a SI-FAR(1)



We fit a SI-FAR(1) and a two state SI-FAR-HMM and compare the fitted FAR(1) kernels for each model. Notice that the SI-FAR-HMM is able to detect state regime behavior, whereas the SI-FAR(1) model is reduced to approximately the average of both the SI-FAR-HMM kernels.

Versus a HMM

We fit a two state HMM model and a two state SI-FAR-HMM and compare Viterbi decoded accuracy of both models; a confusion matrix is presented below. Notice the HMM degenerates to a single state whereas the SI-FAR-HMM maintains a high state decoding accuracy.

Decoded State	True State		
	1	2	Total
1	1052 (491)	13 (574)	1065 (1065)
2	53 (404)	882 (531)	935 (935)
Total	1105 (931)	895 (1105)	2000

Table 1. SI-FAR-HMM decoded states and in brackets HMM decoded states.

References

- [1] Kamila Berhoune and Nawel Bensmain. Sieves estimator of functional autoregressive process. Statistics & Probability Letters, 135:60–69, 2018.
- [2] Ulf Grenander. Abstract Inference. John Wiley & Sons, New York, 1981.